

EXP-1

CODE:

```
Clc;

clear;

close all;

Am = 3;

fm = 800;

Ac = 12;

fc = 12000;

fs = 20*fc;

t = 0:1/fs:4/fm;

m = Am*sin(2*pi*fm*t);

c = Ac*sin(2*pi*fc*t);

mu = Am/Ac;

am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);

dem = abs(hilbert(am));

dem = dem - mean(dem);

N = length(am);

f = linspace(-fs/2, fs/2, N);

AM_FFT = abs(fftshift(fft(am)))/N;

USB = fc + fm;

LSB = fc - fm;

Pc = Ac^2/2;

Psb = (mu^2 * Pc)/4;

figure('Name','AM Modulation','NumberTitle','off'

subplot(3,2,1);plot(t,m);title('Message');grid on;
```

```

subplot(3,2,2);plot(t,c);title('Carrier');grid on;
subplot(3,2,3);plot(t,am);title('AM Signal');grid on;
subplot(3,2,4);plot(t,dem);title('Demodulated');grid on;
subplot(3,2,5);plot(f,AM_FFT);xlim([fc-3*fm fc+3*fm]);title('Sidebands');grid on;

subplot(3,2,6);bar([Pc Psb
Psb]);set(gca,'XTickLabel',{'Carrier','USB','LSB'});title('Power');grid on;

fprintf('mu = %.2f\n', mu);
fprintf('USB = %d Hz\n', USB);
fprintf('LSB = %d Hz\n', LSB);
fprintf('Carrier Power = %.2f W\n', Pc);
fprintf('Each Sideband Power = %.3f W\n', Psb);

```

OUTPUT:

